

Regularized MAR and LSTM in forecasting IV surfaces

References

Jiang, H., Shen, B., Li, Y., & Gao, Z. (2024). Regularized estimation of high-dimensional matrix-variate autoregressive models

Chen, R., Xiao, H., & Yang, D. (2018). Autoregressive models for matrix-valued time series

Li, F.-F., Johnson, J., & Yeung, S. (2017). Lecture 10: Recurrent Neural Networks. CS231n: Convolutional Neural Networks for Visual Recognition (Spring 2017), Stanford University

Options Pricing Overview

- Option: a financial instrument that gives the owner the right, but not the obligation, to buy/sell an underlying asset at a strike price
 - Strike price
 - Time to maturity
- Two types: call options (right to buy) and put options (right to sell)
- European-style options: can only be exercised at expiration

E.g. A European-style, \$90, 30-day expiration, call option gives you the right to buy the underlying stock at \$90 regardless of its market price at expiration

Black-Scholes's Model

$$C = S_0 N(d_1) - K e^{-rT} N(d_2)$$

$$P = K e^{-rT} N(-d_2) - S_0 N(-d_1)$$

where

$$d_1 = \frac{\ln(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

and

S: stock price	r: risk-free interest rate	σ : volatility
K: strike price	T: time to maturity	

Assumptions:

1. The underlying stock does not pay a dividend and never will
2. The option must be European-style
3. Financial markets are efficient
4. No commissions are charged on the trade
5. Interest rates remain constant
6. The underlying stock returns are log-normally distributed

Data Collection

Source:

Bloomberg

Time period:

4/1/2016 - 5/25/2026

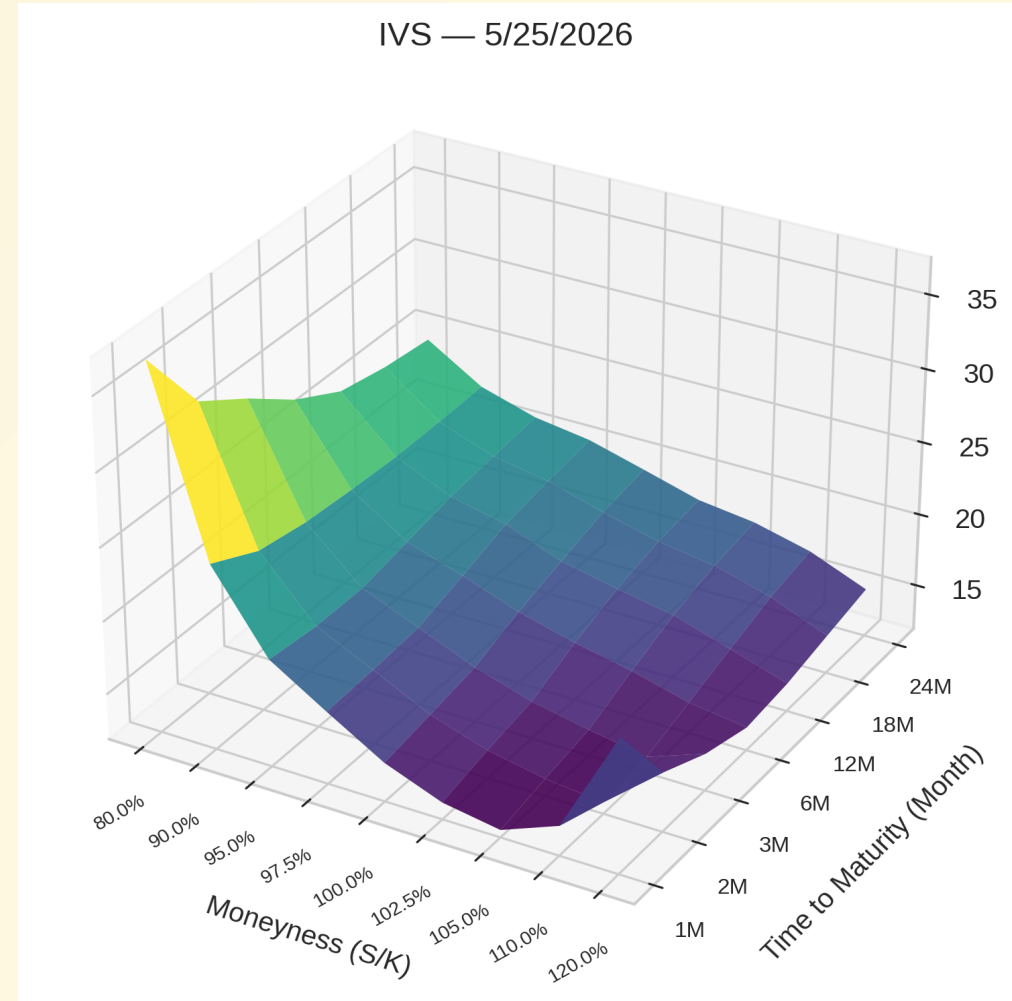
Moneyness (S/K):

80%, 90%, 95%, 97.5%, 100%,
102.5%, 105%, 110%, 120%

Time to maturity:

1M, 2M, 3M, 6M, 12M, 18M, 24M

Each IV surface is a 7x9 matrix



Vector Autoregressive Model (VAR)

Let $X_t \in \mathbb{R}^{m \times n}$ represent the IV matrix at time t

VAR(1):

$$\text{vec}(X_t) = \Phi \text{vec}(X_{t-1}) + e_t$$

where $\text{vec}(\cdot)$ is the vectorization of a matrix by stacking its columns

- Fail to capture relationship between rows and columns
- Number of parameters: $O(m^2n^2)$

Matrix Autoregressive Model (MAR) *(Chen et al., 2020)*

MAR(1):

$$X_t = AX_{t-1}B^\top + E_t$$

- Preserve the matrix structure
- Number of parameters: $O(m^2 + n^2)$

Vectorized representation of MAR(1):

$$\text{vec}(X_t) = (B \otimes A)\text{vec}(X_{t-1}) + \text{vec}(E_t)$$

where $\otimes$ denotes the Kronecker product

MAR Estimation (1): Projection

Project an estimator of the VAR coefficient matrix, denoted as $\hat{\Phi}$, onto the space of Kronecker products under the Frobenius norm

$$(\hat{A}_1, \hat{B}_1) = \arg \min_{A, B} \|\hat{\Phi} - B \otimes A\|_F^2$$

An explicit solution exists, obtained through a singular value decomposition (SVD) of a re-arrangement version of Φ (Van Loan, 2000)

The set of entries of $B \otimes A$ is the same as the set of entries of $\text{vec}(A)\text{vec}(B)'$

Define a re-arrangement operator $\mathcal{G} : \mathbb{R}^{mn \times mn} \rightarrow \mathbb{R}^{m^2 \times n^2}$

$$\mathcal{G}(B \otimes A) = \text{vec}(A)\text{vec}(B)'$$

Given $\|\mathcal{G}(\mathbf{C})\|_F = \|\mathbf{C}\|_F$ and $\mathcal{G}$ is linear

$$\begin{aligned} \min_{\mathbf{A}, \mathbf{B}} \|\hat{\Phi} - \mathbf{B} \otimes \mathbf{A}\|_F^2 &= \min_{\mathbf{A}, \mathbf{B}} \|\mathcal{G}(\hat{\Phi}) - \mathcal{G}(\mathbf{B} \otimes \mathbf{A})\|_F^2 \\ &= \min_{\mathbf{A}, \mathbf{B}} \|\mathcal{G}(\hat{\Phi}) - \text{vec}(\mathbf{A})\text{vec}(\mathbf{B})'\|_F^2 \\ &= \min_{\mathbf{A}, \mathbf{B}} \|\tilde{\Phi} - \text{vec}(\mathbf{A})\text{vec}(\mathbf{B})'\|_F^2, \end{aligned}$$

where $\tilde{\Phi} = \mathcal{G}(\hat{\Phi})$ is the re-arranged $\hat{\Phi}$

Then,

$$\text{vec}(\hat{\mathbf{A}})\text{vec}(\hat{\mathbf{B}})' = d_1 \mathbf{u}_1 \mathbf{v}_1',$$

where d_1 is the largest singular value of $\tilde{\Phi}$, and u_1, v_1 are the corresponding first left and right singular vectors

By converting the vectors into matrices, we obtain the *projection estimators (PROJ)* of A and B , denoted by $\hat{A}_1$ and $\hat{B}_1$, with the normalization that $\|\hat{A}_1\|_F = 1$

MAR Estimation (2): Iterated least square

Assume entries of E_t are i.i.d. with mean zero and constant variance:

$$\min_{A,B} \sum_t \|X_t - AX_{t-1}B'\|_F^2$$

FOCs:

$$\sum_t AX_{t-1}B'BX'_{t-1} - \sum_t X_tBX'_{t-1} = 0$$

$$\sum_t BX'_{t-1}A'AX_{t-1} - \sum_t X'_tAX_{t-1} = 0$$

With probability one, the problem has a unique global minimum, and finitely many local minima

Using the $\hat{A}_1$ and $\hat{B}_1$ (*PROJ*) as the starting values, iteratively update one matrix, $\hat{A}$ or $\hat{B}$, while fixing the other

$$B \leftarrow \left(\sum_t X'_t A X_{t-1} \right) \left(\sum_t X'_{t-1} A' A X_{t-1} \right)^{-1}$$

$$A \leftarrow \left(\sum_t X_t B X'_{t-1} \right) \left(\sum_t X_{t-1} B' B X'_{t-1} \right)^{-1}$$

to obtain $\hat{A}_2$ and $\hat{B}_2$, referred to as *LSE*

Split data: 80% old for train and 20% new for test

```
n_obs = len(Y_var)
split_idx = int(n_obs * 0.8)
Y_train, Y_test = Y_var.iloc[:split_idx], Y_var.iloc[split_idx:]
```

Fit VAR(1)

```
var1_model = VAR(Y_train)
var1_res = var1_model.fit(1)
```

- The coefficient matrix of VAR(1) is a 63x63 matrix (3969 parameters)

Fit VAR(p)

```
varp_model = VAR(Y_train)
varp_aic_res = varp_model.fit(maxlags=15, ic='aic') # p is selected by AIC
varp_bic_res = varp_model.fit(maxlags=15, ic='bic') # p is selected by BIC
```

- Optimal p selected by AIC: 15;
- Optimal p selected by BIC: 1
BIC imposes a greater penalty on complexity

$$\text{AIC} = \ln \left(\frac{\text{SSR}}{T} \right) + (p + 1) \frac{2}{T}, \quad \text{BIC} = \ln \left(\frac{\text{SSR}}{T} \right) + (p + 1) \frac{\ln(T)}{T}$$

Fit MAR(1): converged after 355 iterations where the Frobenius distances are less than 10^{-6}

Regularized MAR with LASSO *(Jiang et al., 2024)*

Rewrite MAR into

$$\text{vec}(Y_t) = ((BY'_{t-1}) \otimes I_m) \text{vec}(A) + \text{vec}(E_t)$$

$$\text{vec}(Y'_t) = ((AY_{t-1}) \otimes I_n) \text{vec}(B) + \text{vec}(E'_t)$$

Denote

$$Z_{t-1} = (BY'_{t-1}) \otimes I_m, \quad \hat{Z}_{t-1} = (\hat{B}Y'_{t-1}) \otimes I_m$$

$$Z_{t-1}^* = (AY_{t-1}) \otimes I_n, \quad \hat{Z}_{t-1}^* = (\hat{A}Y_{t-1}) \otimes I_n$$

Using $\hat{A}_2$ and $\hat{B}_2$ (*LSE*) as the starting values, iteratively solve the optimization to update one matrix, $\hat{A}$ or $\hat{B}$, while fixing the other

$$\hat{\alpha} = \arg \min_{\alpha \in \mathbb{R}^{m^2}} \left\{ \frac{1}{T} \sum_{t=2}^T \|\mathbf{y}_t - \hat{\mathbf{Z}}_{t-1} \alpha\|_2^2 + \lambda_{1,T} \|\alpha\|_1 \right\} \quad (1)$$

$$\hat{\beta} = \arg \min_{\beta \in \mathbb{R}^{n^2}} \left\{ \frac{1}{T} \sum_{t=2}^T \|\mathbf{y}_t^* - \hat{\mathbf{Z}}_{t-1}^* \beta\|_2^2 + \lambda_{2,T} \|\beta\|_1 \right\} \quad (2)$$

The estimated coefficients are consistent under the condition that $\max(m, n)mn/T \rightarrow 0$, and A and B have a finite number of non-zero entries

Fit MAR Lasso

- Use K-Fold Expanding Window CV to select $\lambda_{1,T}$ and $\lambda_{2,T}$

```
lambda_1 = np.logspace(-4, -1, 4) # 4 points between 1e-4 and 1e-1  
lambda_2 = np.logspace(-4, -1, 4) # 4 points between 1e-4 and 1e-1  
tscv = TimeSeriesSplit(n_splits=3) # 3 folds
```

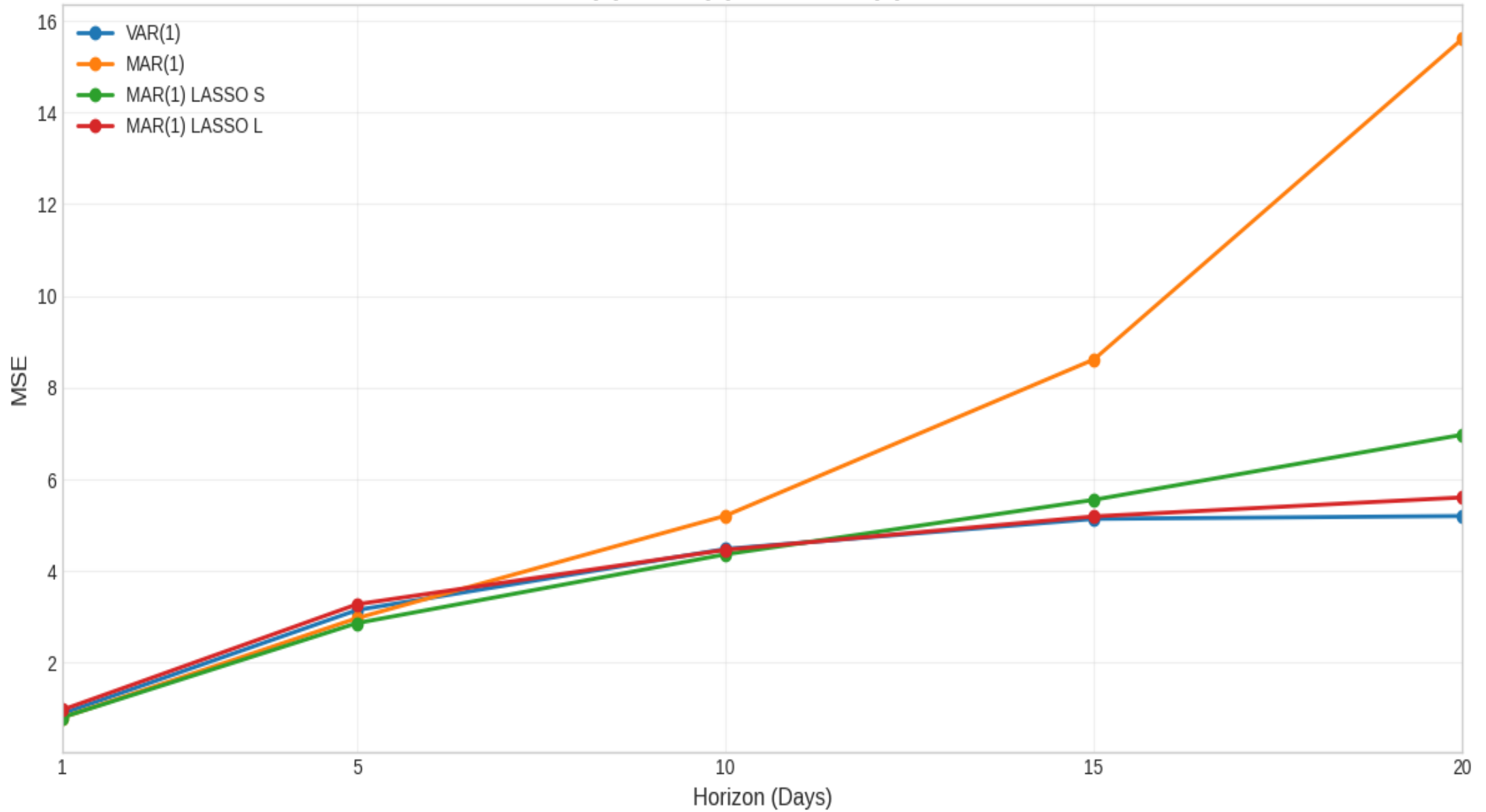
$\lambda_{1,T} = 0.1, \lambda_{2,T} = 0.001, \text{Best MSE} = 1.761308$

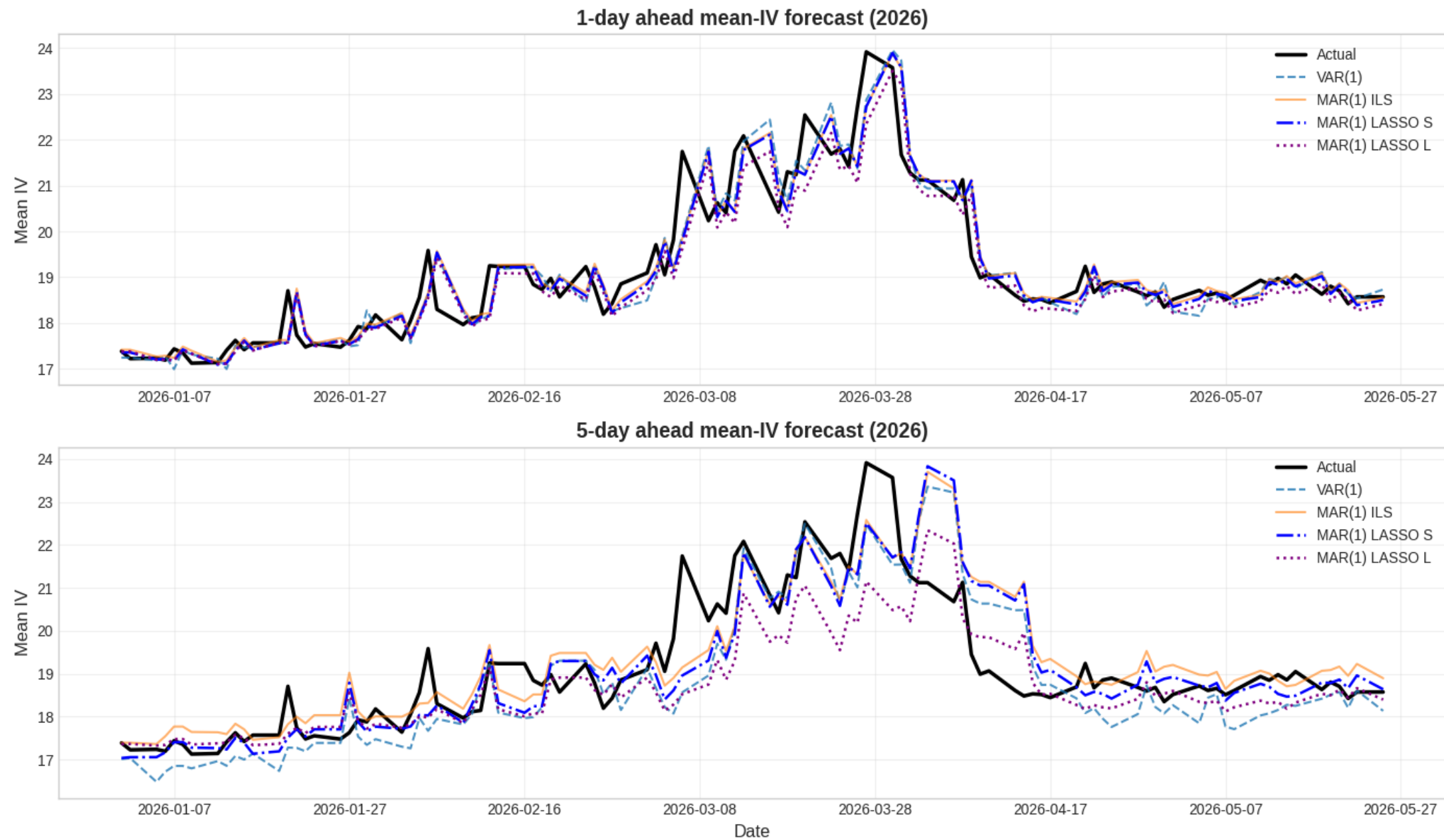
```
lambda_1 = np.logspace(-5, 5, 10) # 10 points between 1e-5 and 1e5  
lambda_2 = np.logspace(-5, 5, 10) # 10 points between 1e-5 and 1e5  
tscv = TimeSeriesSplit(n_splits=5) # 5 folds
```

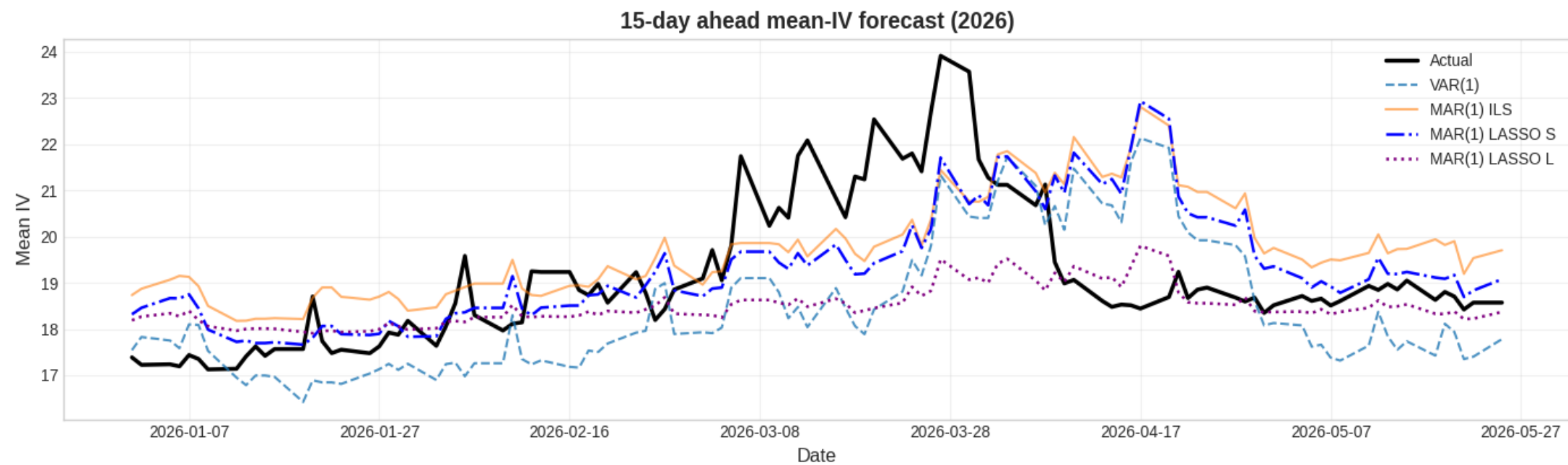
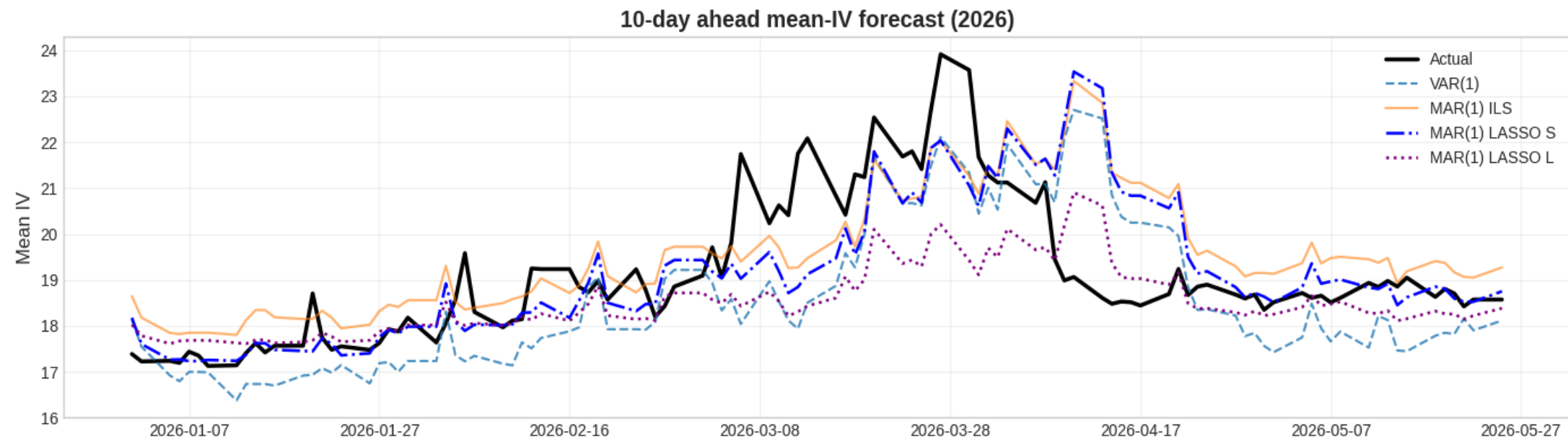
$\lambda_{1,T} = 3.59381, \lambda_{2,T} = 0.00167, \text{Best MSE} = 1.704284$

Horizon (days)	VAR(1)	VAR(15) AIC	MAR(1)	MAR(1) LASSO S	MAR(1) LASSO L
1	0.89	2.45	0.81	0.81	0.98
5	3.15	7.93	2.98	2.87	3.28
10	4.48	9.95	5.21	4.37	4.47
15	5.14	10.43	8.61	5.55	5.19
20	5.20	9.95	15.60	6.97	5.61
25	5.89	10.20	33.35	9.68	5.94
30	6.23	10.77	85.49	13.67	6.01
60	7.37	13.72	74965.18	115.34	5.67
90	6.64	14.42	1.37e8	413.36	5.66

MSE of VAR(1), MAR(1), and MAR(1) Lasso Models







Obs: MAR(1) slightly outperformed VAR(1) for 1-day and 5-day horizons. As the horizon increases, while all the models got worse, MAR(1) completely exploded

- MAR iteratively solves local minimization problems, which more likely yields an eigenvalue greater than 1
- Iterated multi-period forecast raises the eigenvalue to a power equal to the time horizon
- Forecast compounds exponentially and thus MSFE explodes

Obs: There is a positive correlation between the penalty magnitude and time horizon in terms of prediction performance

- MAR(1) LASSO S with smaller penalties improved short-term forecasts but could not avoid long-term explosion
- MAR(1) LASSO L with larger penalties prevented long-term explosion but caused underfitting in the short term

Obs: VAR(15) underperformed against all the first-lagged models

- The Weak Efficient Market Hypothesis (*Eugene Fama, 1970*): today's stock prices reflect all the data of past prices and that no form of technical analysis can aid investors
- The First Fundamental Theorem of Asset Pricing (*Harrison, Kreps, 1979*): a financial market is arbitrage-free if and only if an Equivalent Martingale Measure exists under which the discounted price of all traded assets are martingales.
- IV follows a random walk at least in the short term
- A greater number of lags with no extra information but noise causes overfitting

Recurrent Neural Network

We can process a sequence of vectors $\mathbf{x}$ by applying a **recurrence formula** at every time step:

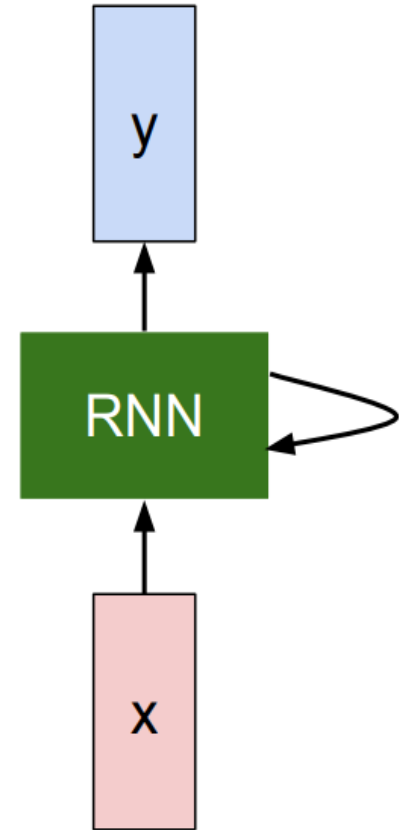
$$\boxed{h_t} = \boxed{f_W}(\boxed{h_{t-1}}, \boxed{x_t})$$

new state

some function with parameters W

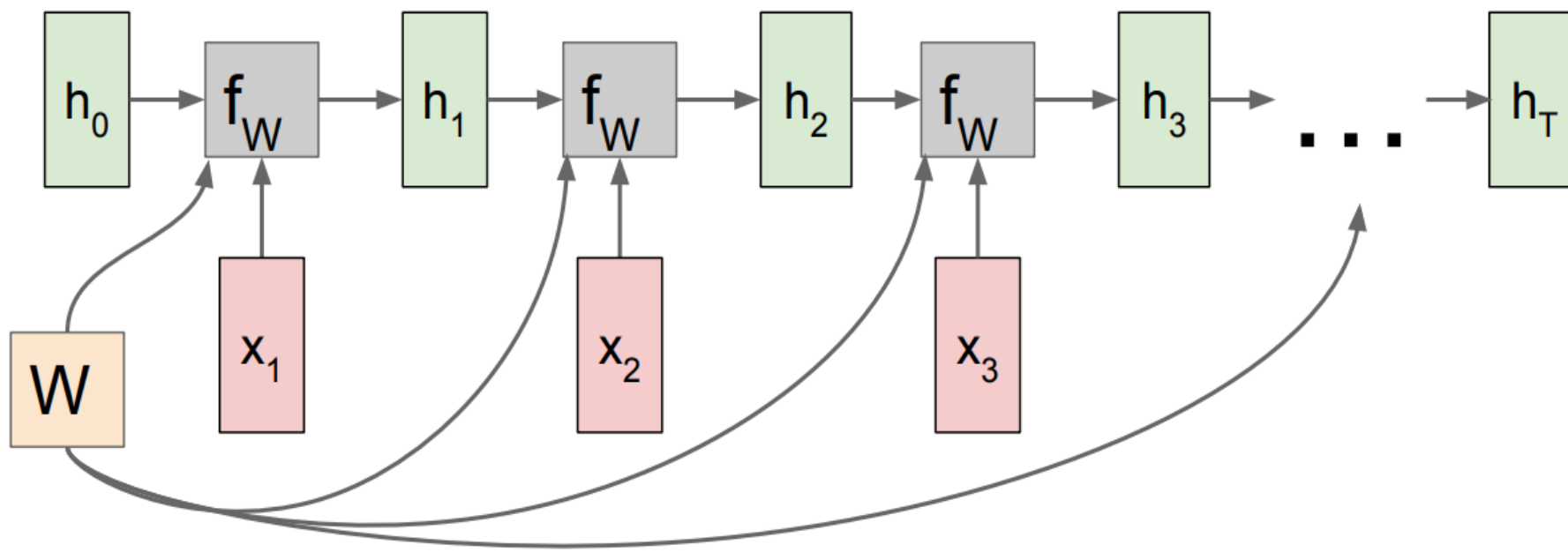
old state

input vector at some time step



RNN: Computational Graph

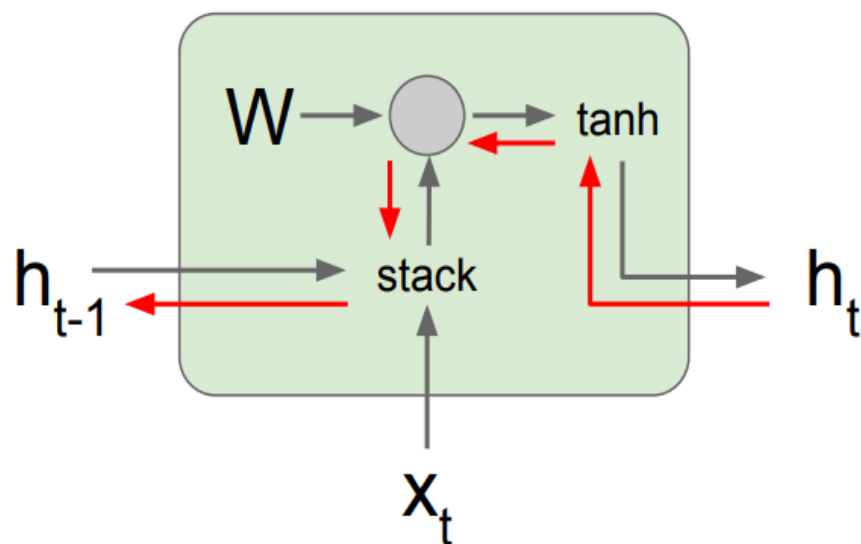
Re-use the same weight matrix at every time-step



Vanilla RNN Gradient Flow

Bengio et al, "Learning long-term dependencies with gradient descent is difficult", IEEE Transactions on Neural Networks, 1994
Pascanu et al, "On the difficulty of training recurrent neural networks", ICML 2013

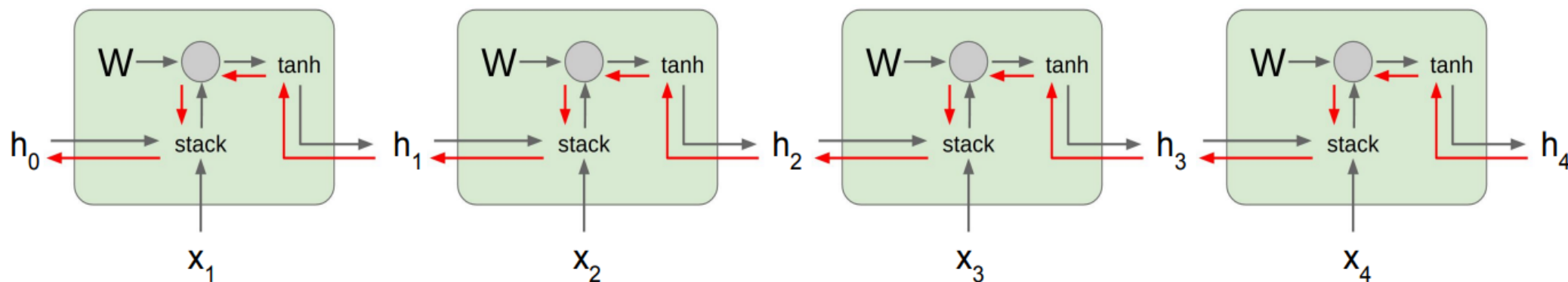
Backpropagation from h_t to h_{t-1} multiplies by W (actually W_{hh}^T)



$$\begin{aligned} h_t &= \tanh(W_{hh}h_{t-1} + W_{hx}x_t) \\ &= \tanh\left(\begin{pmatrix} W_{hh} & W_{hx} \end{pmatrix} \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}\right) \\ &= \tanh\left(W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}\right) \end{aligned}$$

Vanilla RNN Gradient Flow

Bengio et al, "Learning long-term dependencies with gradient descent is difficult", IEEE Transactions on Neural Networks, 1994
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Computing gradient of h_0 involves many factors of W (and repeated $\tanh$)

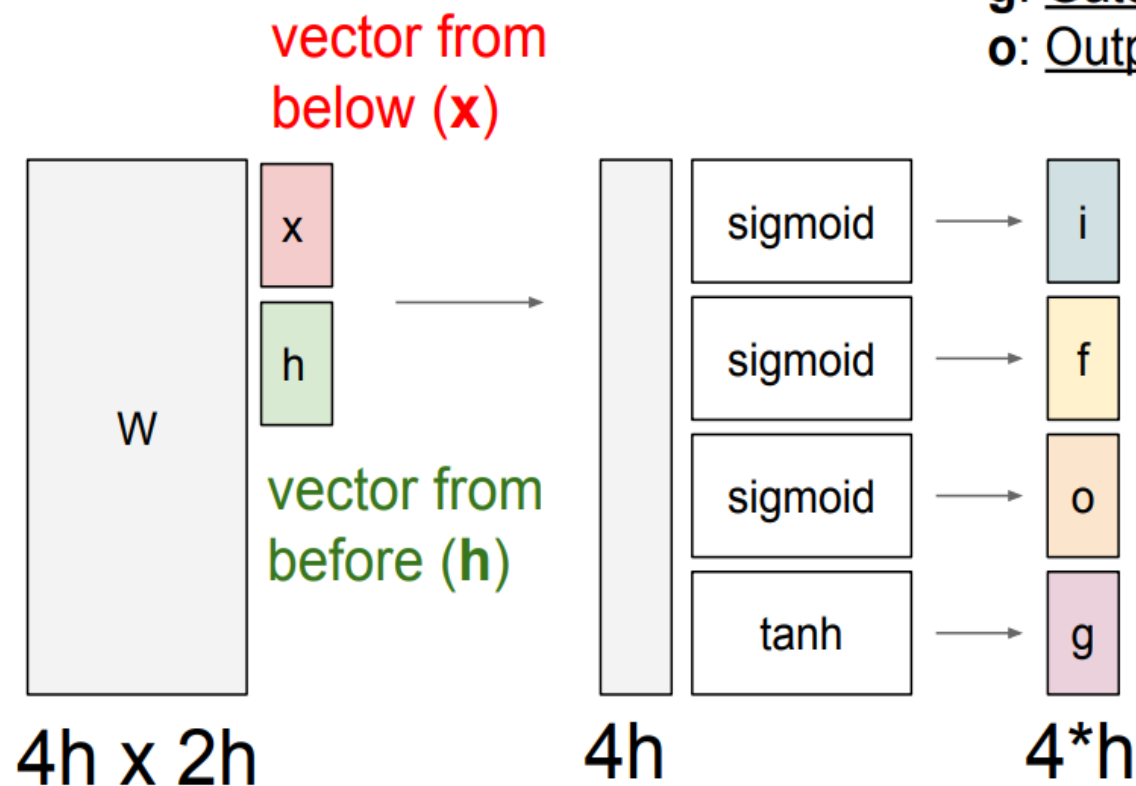
Largest singular value > 1 :
Exploding gradients

Largest singular value < 1 :
Vanishing gradients

Long Short Term Memory (LSTM)

[Hochreiter et al., 1997]

f: Forget gate, Whether to erase cell
i: Input gate, whether to write to cell
g: Gate gate (?), How much to write to cell
o: Output gate, How much to reveal cell



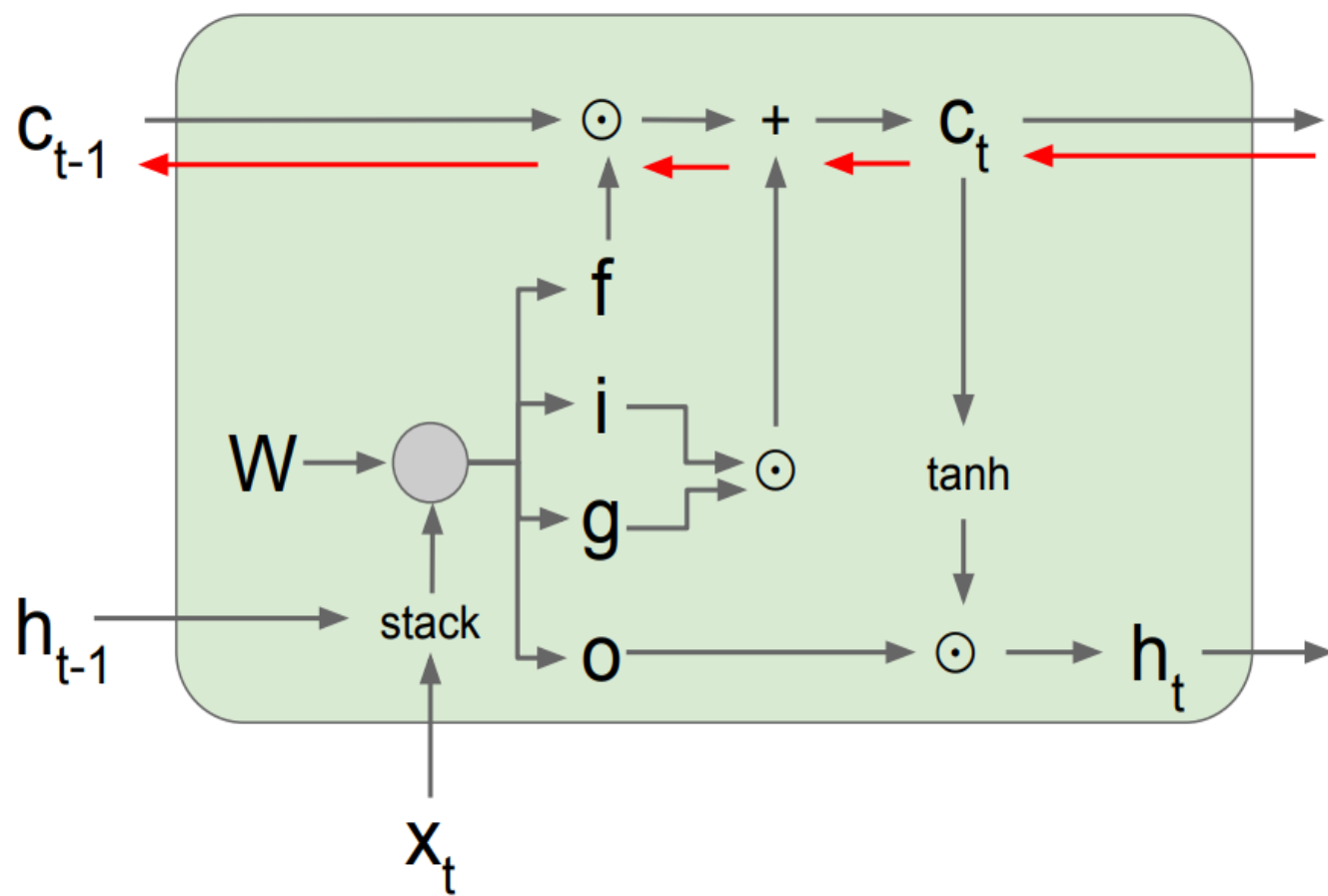
$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$

Long Short Term Memory (LSTM): Gradient Flow

[Hochreiter et al., 1997]



Backpropagation from c_t to c_{t-1} only elementwise multiplication by f , no matrix multiply by W

$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$

Vanilla LSTM vs. Residual LSTM

1. Predict $\Delta_t = y_t - y_{t-1}$ instead of y_t

- Vanilla LSTM:

$$\hat{y}_t = Wh_{t-1}$$

$$\mathcal{L} = ||y_t - Wh_{t-1}||_F^2$$

- Residual LSTM:

$$\hat{y}_t = y_{t-1} + Wh_{t-1}$$

$$\mathcal{L} = ||(y_t - y_{t-1}) - Wh_{t-1}||_F^2$$

Residual LSTM

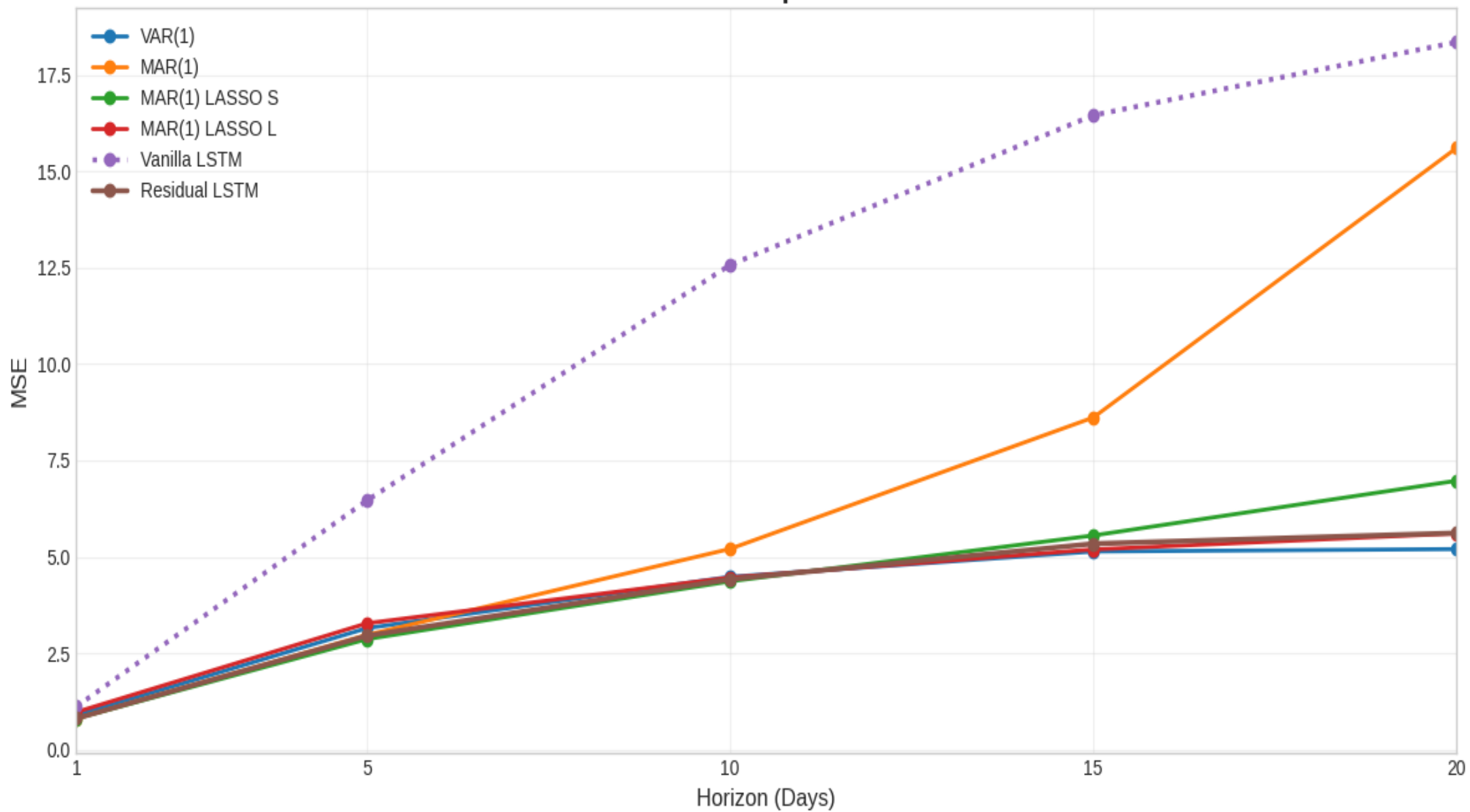
2. Reduce overfitting

- Dropout 20%
- Weight decay (L2 regularization), $\lambda = 10^{-4}$
- Reduce hidden state dimension from 64 to 32

3. Increase epochs: offset the effect of dropout and weight decay

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5	3.15	7.93	2.98	2.87	3.28	7.39	2.87
10	4.48	9.95	5.21	4.37	4.47	16.39	4.18
15	5.14	10.43	8.61	5.55	5.19	22.23	4.92
20	5.20	9.95	15.60	6.97	5.61	26.27	5.15
25	5.89	10.20	33.35	9.68	5.94	30.95	5.74
30	6.23	10.77	85.49	13.67	6.01	33.88	6.05
60	7.37	13.72	74965.18	115.34	5.67	42.39	7.19
90	6.64	14.42	1.37e8	413.36	5.66	46.60	7.17

MSE Comparison



Obs:

Vanilla LSTM suffered from overfitting

Residual LSTM gave a similar result with MAR LASSO S for 1-day and 5-day horizon but remained more stable in the long run

- Overall, Residual LSTM outperformed other models in forecasting IV
- However, predictions may not be comparable due to different response variables
- To extend the study,
 - run econometric models on Δ_t instead of y_t
 - implement ConvLSTM to preserve the matrix structure